Technical Report - Human Resources Candidate Ranking

Duplicate-safe lexical ranking, fold-safe model selection, and recruiter-guided relevance feedback

Selected Stage-1 model: normalized hybrid retrieval (BM25 + word TF-IDF + character TF-IDF) with overlap and HR/intent features, Ridge alpha = 10. Mean grouped-CV NDCG@10 = 0.9628 +/- 0.0621; MAP@10 = 1.000; MRR = 1.000.

1. Objective, data, and labels

The project ranks candidate profiles for two HR-oriented search intents: "aspiring human resources" and "seeking human resources". The output is a relative relevance ordering for shortlist review. It is not a hire/no-hire classifier, a probability of future job performance, or an automated employment decision.

The source contains 104 candidate rows. Exact repetition reduces the effective modelling unit to 53 duplicate-safe profile groups. Repeated copies remain traceable, but all model development and evaluation use the unique profile group as the unit of evidence.

Source rows	104
Unique duplicate-safe profiles	53
Primary text field	job_title
Human relevance target	0-3
Primary ranking metric	NDCG@10
MAP/MRR relevant threshold	grade >= 2

Relevance rubric

Grade 0 is irrelevant to the HR search objective; grade 1 is weak or adjacent; grade 2 is clearly HR-relevant; grade 3 is a strong/direct aspiring or seeking HR match. The unique-profile distribution is 17 / 2 / 28 / 6 for grades 0 / 1 / 2 / 3. After duplicate propagation to all source rows, the row-level distribution is 27 / 10 / 47 / 20.

Data-quality safeguards

Before modelling, the labelled CSV is required to preserve the original five source fields, IDs must be unique, final grades must be complete and restricted to 0-3, every duplicate group must carry one consistent grade, and the grouping procedure must reproduce exactly 53 unique profiles.

Interpretation of the target

The 0-3 target measures textual relevance to the stated recruiter search objective. It deliberately distinguishes direct aspiring/seeking intent from generic HR relevance and from unrelated roles. A high relevance grade does not imply employability, cultural fit, seniority fit, or expected performance.

2. Duplicate handling, text preparation, and fold-safe evaluation

Duplicate-safe grouping

A stable group key is formed from normalized job title, normalized location, and parsed connection count. All copies of one profile are assigned to the same group. Five-fold GroupKFold splits the 53 groups so no repeated copy can appear in both training and validation in one fold. This prevents duplicate leakage from inflating offline ranking metrics.

Text normalization and optional metadata

- Lowercase text and normalize punctuation and whitespace.
- Expand HR to human resources in the normalized representation.
- Normalize entry-level variants to the token entrylevel.
- Preserve intent-bearing terms such as aspiring, seeking, student, internship, and entry-level language.
- Compare surface, normalized, and Porter2-stemmed preprocessing in the broad search.

Connection count is parsed numerically for diagnostics and optional modelling; "500+" becomes 500 for that transformation. The best connection-inclusive configuration scores lower than the selected text-only ranker, so connection count is excluded from the final Stage-1 relevance feature set.

Fold-safe representation fitting

Within each cross-validation fold, word TF-IDF vocabulary/IDF, character TF-IDF vocabulary/IDF, and BM25 corpus statistics are fitted from training titles only. Validation titles do not contribute vocabulary, document frequency, inverse document frequency, or average document length. Held-out titles and the two recruiter queries are transformed or scored using only training-fitted representations.

- Split unique profile groups with GroupKFold.
- Fit word TF-IDF, character TF-IDF, and BM25 on training titles only.
- Build query-profile features separately for training and validation partitions.
- Standardize features and fit Ridge on training rows.
- Compute NDCG, MAP, and MRR on held-out profile groups for each query, then aggregate.

Why the final full-data fit is different

After the cross-validation screen selects a configuration, the final Stage-1 ranker is fitted on all 53 development profiles. At that point the purpose is not to estimate held-out performance; it is to order the complete available candidate pool with the selected method. Final TF-IDF spaces are fitted on candidate titles only, and the recruiter queries are transformed afterward.

Metric roles

NDCG@10 is primary because it uses the full 0-3 relevance grades and rewards putting the strongest candidates near the top. MAP@10 and MRR use grades 2-3 as relevant and act as secondary diagnostics. They saturate more easily on this small dataset, which is why NDCG@10 drives model selection.

3. Representations, features, model search, and quantitative results

Retrieval representations and engineered features

- Word TF-IDF: unigram/bigram cosine similarity for token-level lexical match.
- Character TF-IDF: character n-gram cosine similarity for short titles and lexical variants.
- BM25: probabilistic term matching using training-fold document frequency and length normalization.
- Hybrid: combines BM25, word TF-IDF, and character TF-IDF signals.

Additional interpretable features include Jaccard token overlap, query containment, HR-term overlap, and intent-term overlap. An optional log-transformed connection-count feature is tested only in connection-inclusive bundles. Ridge regression combines the standardized features into a relative ordering score.

192-configuration model-search grid

Preprocessing	surface; normalized; Porter2
Representation	word TF-IDF; word+char TF-IDF; BM25; hybrid
Feature bundles	retrieval; +overlap; +overlap+intent; +connections
Ridge alpha	0.1; 1; 10; 100

The complete screen is 3 x 4 x 4 x 4 = 192 configurations, including 64 Porter2 variants. The selected normalized hybrid + overlap/intent model with Ridge alpha 10 has the highest mean NDCG@10 in the fold-safe screen.

Top model-selection results

Selected normalized hybrid + overlap/intent, alpha 10	0.9628 +/- 0.0621; MAP@10 1.000; MRR 1.000
Normalized word TF-IDF + overlap, alpha 10	0.9622 +/- 0.0634
Normalized word+char TF-IDF + overlap, alpha 10	0.9622 +/- 0.0634
Best model including connection count	0.9562 +/- 0.0649

Direct lexical baselines

Character TF-IDF	0.9535 +/- 0.0582; MAP@10 0.9938; MRR 1.000
BM25	0.9527 +/- 0.0577; MAP@10 0.9909; MRR 1.000
Word TF-IDF	0.9359 +/- 0.0668; MAP@10 0.9874; MRR 1.000

BM25 versus word TF-IDF

Across ten matched observations (5 folds x 2 queries), BM25 minus word TF-IDF has mean NDCG@10 difference +0.0168, paired t-test $p = 0.278$, and 95% CI [-0.0161, +0.0498]. Win/tie/loss is 3/6/1. The observed average favors BM25, but the confidence interval includes zero, so the result does not establish clear BM25 superiority.

The top few text-only configurations are separated by only a few thousandths of NDCG@10. That supports the selected hybrid model for this dataset while also showing that the ranking signal is broadly robust across several transparent lexical formulations. The connection-count feature does not add useful ranking quality.

4. Final ranking, recruiter feedback, limitations, and reproduction

Final Stage-1 ranking

The selected normalized hybrid Ridge model is fitted on all 53 unique profiles. Each profile receives a prediction for each of the two recruiter queries; predictions are normalized and the profile-level Stage-1 score is the maximum across those query-specific predictions.

1. ID 99 1.0000 Seeking Human Resources Position
2. ID 3 0.9770 Aspiring Human Resources Professional
3. ID 97 0.9770 Aspiring Human Resources Professional
4. ID 6 0.9727 Aspiring Human Resources Specialist
5. ID 28 0.9576 Seeking Human Resources Opportunities
6. ID 73 0.8996 Aspiring HR Manager, seeking internship in HR
7. ID 10 0.8482 Seeking HR HRIS and Generalist Positions
8. ID 27 0.8390 Aspiring HR Management student seeking internship
9. ID 79 0.8301 Liberal Arts Major. Aspiring HR Analyst.
10. ID 72 0.8133 Business Management Major and Aspiring HR Manager

These are normalized ranking scores, not hiring probabilities. A score of 1.0000 means the profile receives the highest Stage-1 relevance score within this candidate pool and these two stated search intents.

Stage-2 recruiter feedback

The demonstration maps source ID 3 to its duplicate-safe profile group and treats it as positive feedback; no rejected profile is supplied. Rocchio uses alpha = 1.0, beta = 0.75, gamma = 0.15. The feedback TF-IDF space is fitted on candidate titles only. The final Stage-2 score blends 65% Stage 1 with 35% feedback. Human relevance grades are not silently reused as recruiter feedback.

1. ID 3 Stage-1 2 change +1 Aspiring Human Resources Professional
2. ID 97 Stage-1 3 change +1 Aspiring Human Resources Professional
3. ID 99 Stage-1 1 change -2 Seeking Human Resources Position
4. ID 6 Stage-1 4 change 0 Aspiring Human Resources Specialist
5. ID 28 Stage-1 5 change 0 Seeking Human Resources Opportunities
6. ID 74 Stage-1 11 change +5 Human Resources Professional

The starred profile moves from Stage-1 rank 2 to Stage-2 rank 1. In the full demonstration, the largest upward movement is ID 76 from rank 28 to 20 (+8); ID 1 moves from 27 to 21 (+6); and ID 74 moves from 11 to 6 (+5). The 65/35 blend keeps the general ranker dominant while allowing explicit preferences to alter the current shortlist.

Limitations and responsible use

- Only 53 unique labelled profiles support development and evaluation.
- The 0-3 labels contain human judgement uncertainty; no formal inter-rater agreement estimate is available.
- Job-title text is sparse, lexical retrieval can miss semantic equivalence, and repeated experimentation on a small labelled set can overfit model-selection decisions.
- Offline NDCG, MAP, and MRR do not establish recruiter productivity, hiring quality, or employee performance.
- Omitting protected attributes does not prove fairness; text, location, and network-related variables can still act as proxies.

Reproduction and next steps

Install the pinned environment from requirements.txt, launch Jupyter from the repository root, and run 01_data_representation.ipynb, 02_grouped_cv_model_selection.ipynb, and 03_final_ranking_stage2.ipynb in order. The notebooks read the two CSV files by bare filename and require no helper modules, parent-folder search, build directories, generated-result folders, or external figure files.

The highest-value next steps are to collect more unique labelled profiles, preserve a prospective holdout set, add a second annotator or adjudication process, measure recruiter behavior in a controlled pilot, and compare semantic representations only after the evaluation set is large enough for a meaningful comparison.

Technical conclusion: the evidence supports a transparent normalized hybrid Ridge ranker with alpha 10 for this dataset. It achieves the best mean grouped-CV NDCG@10 in the cleaned 192-configuration search, excludes connection count, and supports controlled recruiter reranking through explicit feedback. The main constraint is the amount of unique labelled data, not a lack of model complexity.